SAFETY DATA SHEET

1. Identification

Product identifier	**Rapidur B 181**
Recommended use	Foundry binder
Recommended restrictions	Not available
Manufacturer/Importer/Supplier/Distributer information	
Manufacturer:	Mancuso Chemicals Limited 5725 Progress Street Niagara Falls, ON, Canada L2G 0C1
Company telephone	1-905-357-3626
Emergency telephone	1-800-567-7455 (Terrapure Environmental)
Supplier	Refer to Manufacturer

2. Hazard(s) identification

Physical hazards	Flammable liquid	Category 4
Health hazards	Acute toxicity, Inhalation	Category 4
	Acute toxicity, Oral	Category 4
	Serious eye damage/eye irritation	Category 2A
	Skin corrosion/irritation	Category 2
	Skin sensitization	Category 1
	Respiratory sensitization	Category 1
	Specific target organ toxicity - single exposure	Category 3, (Respiratory system)
	Specific target organ toxicity - repeated exposure	Category 1, (Respiratory tract)

Label elements

Signal word	Danger
Hazard statement	Combustible liquid. May be fatal if swallowed and enters airways. Causes skin irritation. May cause an allergic skin reaction. Causes serious eye irritation. Harmful if inhaled. May cause allergy or asthma symptoms or breathing difficulties if inhaled. May cause respiratory irritation. Suspected of causing cancer. May cause damage to organs through prolonged or repeated exposure.
Precautionary statement	
Prevention	Obtain special instructions before use. Do not handle until all safety precautions have been read and understood. Keep away from heat/sparks/open flames/hot surfaces. - No smoking. Do not breathe mist or vapor. Ground/bond container and receiving equipment. Use only non-sparking tools. Wash thoroughly after handling. Do not eat, drink or smoke when using this product. Use only outdoors or in a well-ventilated area. Wear personal protective equipment as required.
Response	If swallowed: Immediately call a poison center/doctor. Rinse mouth. Do NOT induce vomiting. If on skin: Wash with plenty of water. If inhaled: Remove person to fresh air and keep comfortable for breathing. If in eyes: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. If eye irritation persists: Get medical advice/attention. Call a poison center/doctor. Take off immediately all contaminated clothing and wash it before reuse. In case of fire: use dry sand, dry chemical or alcohol-resistant foam for extinction.
Storage	Store in a well-ventilated place. Keep container tightly closed. Keep cool. Store locked up.
Disposal	Dispose of contents/container in accordance with local/regional/national/international regulations.
Hazards not otherwise classified	None known.
Supplemental information	None.

3. Composition/information on ingredients

Chemical name	Comnmon name and synonyms	CAS Number	% w/w
Polymeric MDI		9016-87-9	22-52
Diphenylmethane-diisocyanate (mixed isomers)		26447-40-5	22-52
Solvent Naptha (Petroleum), Heavy Aromatic		64742-94-5	15-30

4. First-aid measures

Inhalation	Remove victim to fresh air and keep at rest in a position comfortable for breathing. Oxygen or artificial respiration if needed. Do not use mouth-to-mouth method if victim inhaled the substance. Induce artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Call a POISON CENTER or doctor/physician.
Skin contact	Take off immediately all contaminated clothing. Wash off with soap and water. Get medical advice/attention if you feel unwell. Get medical attention if irritation develops and persists. Wash contaminated clothing before reuse.
Eye contact	Immediately flush eyes with plenty of water for at least 15 minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Get medical attention if irritation develops and persists.
Ingestion	Call a physician or poison control center immediately. Rinse mouth. Do not induce vomiting. If vomiting occurs, keep head low so that stomach content doesn't get into the lungs. Do not use mouth-to-mouth method if victim ingested the substance. Induce artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device.
Most important symptoms/effects, acute and delayed	Aspiration may cause pulmonary edema and pneumonitis. Severe eye irritation. Symptoms may include stinging, tearing, redness, swelling, and blurred vision. May cause respiratory irritation. Prolonged exposure may cause chronic effects.
Indication of immediate medical attention and special treatment needed	Provide general supportive measures and treat symptomatically. Keep victim warm. Keep victim under observation. Symptoms may be delayed.
General information	Take off immediately all contaminated clothing. IF exposed or concerned: Get medical advice/attention. If you feel unwell, seek medical advice (show the label where possible). Ensure that medical personnel are aware of the material(s) involved, and take precautions to protect themselves. Show this safety data sheet to the doctor in attendance.

5. Fire-fighting measures

Suitable extinguishing media	Alcohol resistant foam. Dry chemical powder. Carbon dioxide (CO_2), or high expansion (proteinic) foam, water spray or fog.
Unsuitable extinguishing media	Do not use water jet as an extinguisher, as this will spread the fire.
Specific hazards arising from the chemical	Do not allow run-off from firefighting to enter drains or water courses.
Special protective equipment and precautions for firefighters	Self-contained breathing apparatus and full protective clothing must be worn in case of fire.
Fire fighting equipment/instructions	In case of fire and/or explosion do not breathe fumes. Use water spray to cool fully closed containers.
Specific methods	Use standard firefighting procedures and consider the hazards of other involved materials.
General fire hazards	Flammable liquid
Hazardous combustion products	Toxic fumes, gases or vapours may evolve on burning. Carbon oxides. Hydrocarbons. Do not mix with strong acids or mineral acids.

6. Accidental release measures

Personal precautions, protective equipment and emergency procedures	Keep unnecessary personnel away. Keep people away from and upwind of spill/leak. Eliminate all ignition sources (no smoking, flares, sparks, or flames in immediate area). Wear appropriate protective equipment and clothing during clean-up. Do not breathe mist or vapor. Do not touch damaged containers or spilled material unless wearing appropriate protective clothing. Ensure adequate ventilation. Local authorities should be advised if significant spillages cannot be contained. For personal protection, see section 8 of the SDS.
Methods and materials for containment and cleaning up	Eliminate all ignition sources (no smoking, flares, sparks, or flames in immediate area). Keep combustibles (wood, paper, oil, etc.) away from spilled material. This product is miscible in water. Large Spills: Stop the flow of material, if this is without risk. Dike the spilled material, where this is possible. Cover with plastic sheet to prevent spreading. Use a non-combustible material like vermiculite, sand or earth to soak up the product and place into a container for later disposal.Following product recovery, flush area with water. Small Spills: Absorb with earth, sand or other non-combustible material and transfer to containers for later disposal. Wipe up with absorbent material (e.g. cloth, fleece). Clean surface thoroughly to remove residual contamination. Never return spills to original containers for re-use. For waste disposal, see section 13 of the SDS.
Environmental precautions	Avoid discharge into drains, water courses or onto the ground.

7. Handling and storage

Precautions for safe handling	Obtain special instructions before use. Do not handle until all safety precautions have been read and understood. Keep away from open flames, hot surfaces and sources of ignition. Do not breathe mist or vapor. Avoid contact with eyes, skin, and clothing. Avoid prolonged exposure. Do not taste or swallow. When using, do not eat, drink or smoke. Should be handled in closed systems, if possible. Use only outdoors or in a well-ventilated area. Wear appropriate personal protective equipment. Wash hands thoroughly after handling. Wash contaminated clothing before reuse. Observe good industrial hygiene practices.
Conditions for safe storage, including any incompatibilities	Store locked up. Keep away from heat, sparks and open flame. Store in a cool, dry place out of direct sunlight. Store in original tightly closed container. Store in a well-ventilated place. Keep in an area equipped with sprinklers. Store away from incompatible materials (see Section 10 of the SDS). Once opened, driers attached to the totes or drums are strongly recommended to keep the product dry and free of moisture. Moisture will react with the product to form crusting and sludge.

8. Exposure controls/personal protection

Occupational exposure limits

US OSHA Table Z-1 Limits for Air Contaminants (29 CFR 1910.1000)

Components	Type	Value
Polymeric MDI (CAS 9016-87-9)	Ceiling	0.2 mg/m3
		0.02 ppm
Naphthalene (CAS 91-20-3)	TWA	10 ppm

US. ACGIH Threshold Limit Values

Components	Type	Value
Polymeric MDI (CAS 9016-87-9)	TWA	0.005 ppm

US. NIOSH: Pocket Guide to Chemical Hazards

Components	Type	Value
Polymeric MDI (CAS 9016-87-9)	Ceiling	0.2 mg/m3
	TWA	0.05 mg/m3
		0.005 ppm
1,2,4-Trimethylbenzene (CAS 93-63-6)	TWA	25 ppm
1,3,5-Trimethylbenzene (CAS 108-67-8)	TWA	25 ppm

Appropriate engineering controls	Good general ventilation (typically 10 air changes per hour) should be used. Ventilation rates should be matched to conditions. If applicable, use process enclosures, local exhaust ventilation, or other engineering controls to maintain airborne levels below recommended exposure limits. If exposure limits have not been established, maintain airborne levels to an acceptable level. Provide eyewash station
Individual protection measures, such as personal protective equipment	
Eye/face protection	Chemical respirator with organic vapor cartridge and full facepiece.
Skin protection	
Hand protection	Wear appropriate chemical-resistant gloves.
Other	Wear appropriate chemical-resistant clothing. Use of an impervious apron is recommended.
Respiratory protection	Chemical respirator with organic vapor cartridge and full facepiece
Thermal hazards	Wear appropriate thermal protective clothing, when necessary.
General hygiene considerations	When using do not smoke. Keep away from food and drink. Always observe good personal hygiene measures, such as washing after handling the material and before eating, drinking, and/or smoking. Routinely wash work clothing and protective equipment to remove contaminants.

9. Physical and chemical properties

Appearance	
Physical state	Liquid
Form	Liquid
Color	Dark amber
Odor	Mild typical
Odor Threshold	Not Available
pH	Not Available (Non Aqueous solution)
Melting point/Freezing point	Not Applicable
Initial boiling point	Initial for MDI component >204 ºC (decomposes)
Flash point	64 °C (closed cup)
Evaporation rate	Not Available
Flammability (solid, gas)	Not Available
Lower explosive limit	1.0 for aromatic hydrocarbons
Upper explosive limit	6.0 for aromatic hydrocarbons
Vapor pressure	Not Available
Vapor density	Not Available
Specific gravity	1.06 - 1.10
Solubility(ies)	
Water solubility	Insoluble. Will react to produce CO_2
Partition coefficient (water/octanol)	Not Available
Auto-ignition temperature	Not determined (> 300 ° C)
Decomposition temperature	Not Available
Viscosity	Not Available

10. Stability and reactivity

Reactivity	The product is stable and non-reactive under normal conditions of use, storage and transport.
Chemical stability	Material is stable under normal conditions.
Possibility of hazardous reactions	Hazardous Polymerisation**:** May occur with water, strong acids and bases or extremely high temperatures.
Conditions to avoid	Avoid heat, sparks, open flames and other ignition sources.
Incompatible materials	Alcohols, water, amines, acids, strong bases, aluminum and copper alloys.
Hazardous decomposition products	May form carbon dioxide, monoxide, oxides of nitrogen (NO_x), hydrocarbons and aromatics.

11. Toxicological information

Information on likely routes of exposure

Inhalation	Harmful if inhaled
Skin contact	May cause allergic skin reaction
Eye contact	Causes serious eye irritation
Ingestion	Harmful if swallowed

Components	**Species**	**Test Results**
Polymeric MDI		
Acute		
Dermal LD50	Rabbit	>6,200 mg/kg
Inhalation LC50	Rat	490 mg/kg, 4 hrs, aerosol
Oral LD50	Rat	>10,000 mg/kg
Diphenylmethane-diisocynate (mixed isomers)		
Acute		
Dermal LD50	Rabbit	> 1,000 mg/kg
Inhalation LC50	Mouse	> 2,200 mg/kg
Oral LD50	Rat	370 mg/kg, 4 hrs, aerosol
Solvent Naphtha (Petroleum) Heavy Aromatic		
Acute		
Dermal LD50	Rabbit	>3,000 mg/kg
Inhalation LC50	Rat	3,000 mg/kg
Oral LD50	Rat	> 3,800 mg/m3, 4 hr

Skin corrosion/irritation	Causes skin irritation
Serious eye damage/eye irritation	Causes eye irritation
Respiratory or skin sensitization	
Respiratory sensitization	Not a respiratory sensitizer
Skin sensitization	May cause an allergic skin reaction
Germ cell mutagenicity	Classification not possible
Carcinogenicity	Classification not possible
Reproductive toxicity	Classification not possible
Specific organ toxicity	
Single exposure	Classification not possible
Repeated exposure	Classification not possible
Aspiration toxicity	Harmful if inhaled

12. Ecological information

Ecotoxicity	The product is not classified as environmentally hazardous. However, this does not exclude the possibility that large or frequent spills can have a harmful or damaging effect on the environment.
Hazards to aquatic environment (acute)	Not available.
Hazards to aquatic environment (long term)	Not available.
Persistence and degradability	Not available

13. Disposal considerations

Disposal instructions	Collect and reclaim or dispose in sealed containers at licensed waste disposal site. Dispose of contents/container in accordance with local/regional/national/international regulations.
Local disposal regulations	Dispose in accordance with all applicable regulations.
Hazardous waste code	The waste code should be assigned in discussion between the user, the producer and the waste disposal company.
Waste from residues / unused products	Dispose of in accordance with local regulations. Empty containers or liners may retain some product residues. This material and its container must be disposed of in a safe manner (see: Disposal instructions).
Contaminated packaging	Empty containers should be taken to an approved waste handling site for recycling or disposal. Since emptied containers may retain product residue, follow label warnings even after container is emptied.

14. Transport information

DOT	
UN Number	
Proper shipping name	Not regulated
Transport hazard class(es)	
Class	
Subsidiary risk	-
Label(s)	
Packing group	
IATA	
UN Number	
Proper shipping name	Not regulated
Transport hazard class(es)	
Class	
Subsidiary risk	-
Packing group	
Environmental hazards	
IMDG	
UN Number	
Proper shipping name	Not regulated
Transport hazard class(es)	
Class	
Subsidiary risk	-
Packing group	
Environmental hazards	
Marine pollutant	

15. Regulatory information

US federal regulations	Not listed
TSCA Section 12(b) Export Notification (40 CFR 707, Subpt. D)	All hazardous products listed
CERCLA Hazardous Substance List (40 CFR 302.4)	Diphenylmethane-diisocyanate (mixed isomers), 5,000 lbs. (2270kg)
OSHA Specifically Regulated Substances (29 CFR 1910.1001-1050	Not listed
Superfund Amendments and Reauthorization Act of 1986 (SARA)	Sec. 313: Diphenylmethane-diisocyanate (mixed isomers), 1% de minimis

16. Other information, including date of preparation or last revision

Issue date	July 28, 2017
Revision	2